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Research on Educational Equity Evaluation and Optimization Strategy Based on Computer Algorithm

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***Abstract*—This paper constructs a comprehensive evaluation model of educational equity, which integrates multi-source data such as students' academic performance, allocation of educational resources, teachers' strength, school infrastructure and students' socio-economic background, and fully reflects the multi-dimensional characteristics of educational equity. By designing a multi-dimensional evaluation index system, adopting data preprocessing and standardization methods, and combining expert scoring and machine learning algorithm to determine the weight, the comprehensive evaluation of educational fairness is realized. In addition, the study also designed an optimization strategy recommendation algorithm, combined with association rule mining technology, analyzed the relationship between educational fairness and various influencing factors, and provided targeted optimization suggestions. The empirical study collected detailed data of 100 schools through questionnaires and interviews. The analysis found that there were significant differences in the enrollment opportunities between urban and rural areas and between regions, the distribution of educational resources was uneven, high-quality teachers were mainly concentrated in key schools in cities, the infrastructure of rural schools was generally backward, and the distribution of students' academic achievements with different socio-economic backgrounds was significantly different. The educational equity evaluation model and optimization strategy recommendation algorithm based on computer algorithm proposed in this study provide new ideas and methods for scientific evaluation and effective optimization of educational equity, which is of great significance to promoting the continuous improvement of educational equity.**

Keywords—computer algorithm, educational equity evaluation, optimization strategy, evaluation model

I. Introduction

Educational unfairness is not only reflected in the regional distribution of educational resources and the uneven distribution between urban and rural areas, but also profoundly affects the educational opportunities and achievements of different social and economic backgrounds, genders, races and other groups. With the rapid development of information technology, especially the wide application of computer algorithms in the field of education, it provides new possibilities for scientifically and objectively evaluating educational fairness and exploring effective optimization strategies [1]. The purpose of this study is to accurately evaluate educational equity by using advanced computer algorithm technology, and then put forward targeted optimization strategies in order to provide scientific basis and practical path for promoting educational equity, improving educational quality and narrowing educational gap. This is not only of great significance for building a harmonious society and achieving the goal of sustainable development, but also an inevitable requirement for responding to the global educational equity initiative and promoting the process of educational modernization [2].

With the rise of big data, AI, cloud computing and other technologies, the field of education is undergoing unprecedented changes. As the core of these technologies, computer algorithms have shown great potential in many aspects, such as educational evaluation, personalized learning, and teaching resource allocation [3-4]. For example, analyzing students' learning behavior data through machine learning algorithm can provide students with personalized learning paths and resource recommendations; Using big data analysis technology, we can reveal the unbalanced distribution of educational resources and provide decision support for policy makers.

However, despite the convenience brought by technology, how to ensure the educational fairness in the process of technology application is still an urgent problem to be solved. Traditional education evaluation often relies on manual judgment and empirical judgment, and is easily influenced by subjective factors, so it is difficult to guarantee the objectivity and accuracy of evaluation [5]. At the same time, the allocation of educational resources is often based on historical data and empirical decisions, lacking scientific basis and dynamic adjustment mechanism, which easily leads to uneven resource allocation.

Therefore, this study intends to introduce computer algorithms into the evaluation and optimization of educational equity, and realize accurate measurement and effective improvement of educational equity by constructing scientific evaluation models and optimization strategies. This is not only an innovative attempt to the existing education evaluation system, but also an important way to promote education equity and improve education quality by using technical means.

II. Model construction

Study and construct a comprehensive evaluation model of educational equity (Figure 1), which integrates multi-source data, including students' academic performance, allocation of educational resources, teachers, school infrastructure, students' socio-economic background, etc., to fully reflect the multi-dimensional characteristics of educational equity [6-7]. Considering the complexity and large-scale of educational data, it is decided to adopt a

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distributed processing architecture based on cloud computing to ensure the efficiency and scalability of data processing.

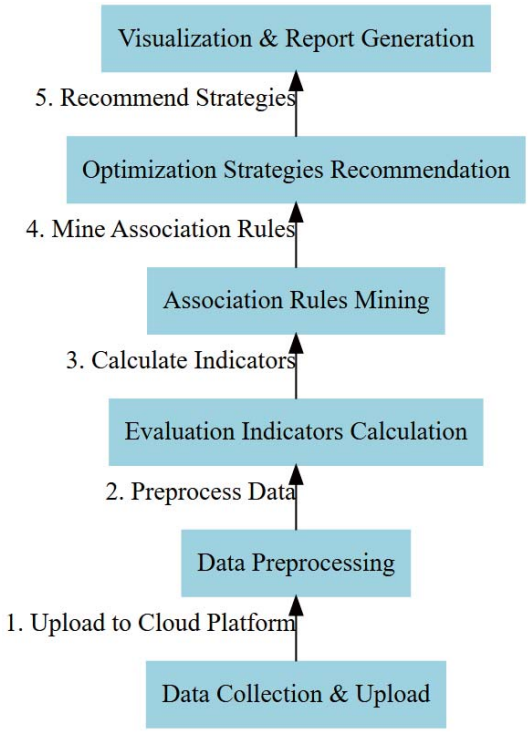


Fig. 1. Comprehensive framework of educational equity evaluation model

A. Design of educational equity evaluation algorithm

In order to comprehensively evaluate educational equity, a multi-dimensional evaluation index system covering three categories: educational opportunity equity, educational process equity and educational outcome equity is constructed, as shown in Figure 2. Among them, the equality of educational opportunities involves the equality of admission opportunities and the balance of educational resources distribution; The education process pays attention to the balance of teachers, the perfection of school infrastructure, the curriculum and teaching quality; The fairness of educational results measures the final results of students from different backgrounds through the distribution of academic achievements, graduation rate and enrollment rate.

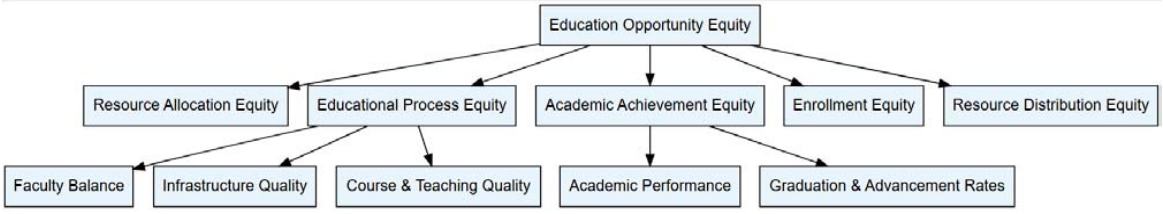


Fig. 2. Multi-dimensional evaluation index system

Because of the different dimensions and value ranges of each index, data preprocessing and standardization are needed. Use the following formula for standardization:

$$x' = \frac{x - \mu}{\sigma} \tag{1}$$

Where x is the original data, μ is the average value of the index and σ is the standard deviation. The standardized data x' will have a unified dimension and value range.

The weight distribution is based on the combination of expert scoring and machine learning. First, experts in the field of education are invited to score each index and get the initial weight [8-9]. Then, the machine learning algorithm is used to optimize the initial weight, and the final weight vector $w = [w_1, w_2, \cdots, w_n]$ is obtained, where w_i is the weight of the i index.

Comprehensive evaluation is carried out by using the method of weighted summation. Let the standardized value of each index be $x' = [x_1', x_2', \cdots, x_n']$, then the comprehensive score S can be calculated by the following formula:

$$S = \sum_{i=1}^{n} w_i \cdot x_i' \tag{2}$$

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Where n is the number of indicators, w_i is the weight of the i indicator, and $x_i^{'}$ is the standardized value of the i indicator.

B. Optimization strategy recommendation algorithm

Research and design an optimization strategy recommendation algorithm. The algorithm combines the association rule mining technology to analyze the correlation between educational fairness and various influencing factors, and provides targeted optimization suggestions for different regions, schools or student groups according to these correlations.

Firstly, the association rules mining technology is used to analyze the relationship between educational equity and various influencing factors (such as the allocation of educational resources, teachers, school infrastructure, students' socio-economic background, etc.). The mined association rules are expressed in the form of "if ... then ...", for example, "If the per capita education expenditure in a certain area is low, the standard deviation of academic performance in that area is large".

Apriori algorithm is used to mine the association rules between educational resource allocation and student performance [10]. Set the minimum support min_sup and the minimum confidence min_conf . Find frequent itemsets L :

$$L = \left\{ itemset : support\left(itemset \right) \geq min_sup \right\} \quad (3)$$

For each frequent itemset $l \in L$, all possible rules $a \Rightarrow b$ are generated, where $a \cup b = l$:

$$conf\left(a \Rightarrow b \right) = \frac{support\left(l \right)}{support\left(a \right)} \quad (4)$$

Keep the rules that satisfy $conf\left(a \Rightarrow b \right) \geq min_conf$.

Based on the matched association rules, personalized suggestions for optimizing educational resources are provided for different regions, schools or student groups. The recommended strategies include increasing educational investment in specific areas, optimizing the allocation of teachers and improving school infrastructure.

For each area/school i to be optimized, the educational fairness score $score_i$ is calculated. According to the association rules and $score_i$, the optimization strategy S_i is determined:

$$S_i = \left\{ action_j : rule_j \in R \wedge condition_j\left(score_i \right) \right\} \quad (5)$$

Where R is the association rule set, $action_j$ is the recommended optimization action, and $condition_j$ is the applicable condition of the rule.

C. Data processing and analysis methods

Obtain education-related data from education departments, schools, statistical institutions and other sources, including students' grades, allocation of educational resources, teachers, school infrastructure, students' socio-economic background, etc. Use Python to clean the collected data and remove duplicates, missing or abnormal values. Standardize the data to ensure that data from different sources and formats can be compared and analyzed uniformly.

Upload the preprocessed data to the cloud computing platform Hadoop HDFS. Using the distributed storage capability of cloud computing platform, large-scale educational data can be stored efficiently. Using Hadoop's MapReduce programming model, write distributed computing tasks, and process and analyze large-scale data. Finally, combined with Python, R and other data analysis languages, as well as Tableau, Power BI and other data visualization tools, the processed data are deeply analyzed to reveal the current situation, trends and existing problems of educational equity.

III. Empirical research and analysis

A. Data source

Through questionnaires and interviews, this paper collected detailed data on the allocation of educational resources, curriculum and teaching quality from 100 schools in different regions and different classes (including 30 primary schools, 40 junior high schools and 30 senior high schools). Among them, each school selected an average of 5 teachers and 10 students as the survey objects, and obtained a total of 500 teachers' questionnaires and 1000 students' questionnaires; At the same time, in-depth interviews were conducted with 20 schools to obtain more comprehensive information. The collected data are integrated, and the data are cleaned and preprocessed by using the parallel processing ability of the cloud computing platform. Store the processed data on the cloud computing platform and establish a data warehouse for subsequent data analysis and processing.

B. Empirical analysis

Using the comprehensive education equity evaluation model, the collected data are processed and analyzed. The model combines multi-source data, including students' academic performance, allocation of educational resources, teachers, school infrastructure, students' socio-economic background, etc., to fully reflect the multi-dimensional characteristics of educational equity.

By analyzing the enrollment rate of students from different regions and socio-economic backgrounds, it is found that there are significant differences in enrollment opportunities between urban and rural areas and between regions. By evaluating the distribution of educational resources (such as books, experimental equipment, information equipment, etc.) between urban and rural areas, regions and schools, it is found that the distribution of resources is uneven, and the resources of urban schools are obviously better than those of rural schools. As can be seen from Figure 3, urban schools have 60% of educational resources, while rural schools have 40%. This distribution ratio reflects the unbalanced distribution of resources between urban schools and rural schools.

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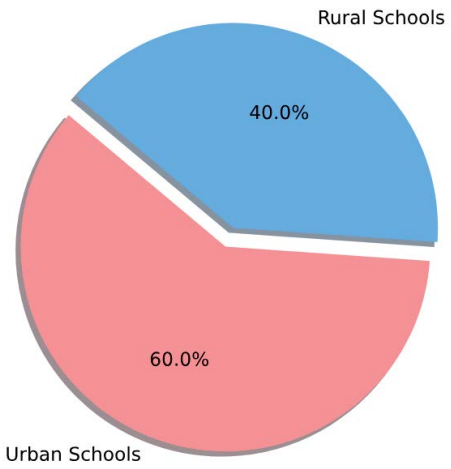


Fig. 3. Distribution ratio of educational resources between urban schools and rural schools

By analyzing the distribution of teachers' academic qualifications, professional titles and teaching experience among different schools, it is found that high-quality teachers are mainly concentrated in urban key schools. Assessing the perfection of school classrooms, laboratories, sports facilities and other infrastructure, it is found that the infrastructure of rural schools is generally backward. Through student questionnaires and teaching quality evaluation, it is found that there are significant differences in curriculum and teaching quality among different schools.

Figure 4 shows the distribution of academic achievements of students from different socio-economic backgrounds (urban and rural). In the picture, blue represents the achievement distribution of urban students, and green represents the achievement distribution of rural students. As can be seen from the figure, there are significant differences in the distribution of grades, and the grades of urban students are generally higher than those of rural students.

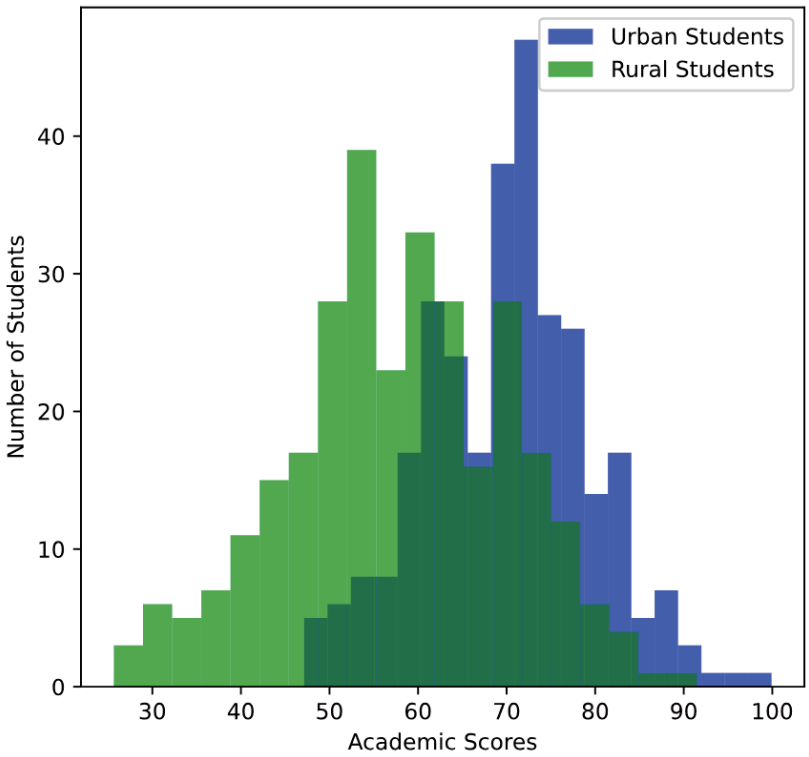


Fig. 4. Distribution of academic achievements of students with different socio-economic backgrounds

By evaluating the graduation rate and enrollment rate of students from different schools and backgrounds, it is found that the graduation rate and enrollment rate of schools with rich educational resources are higher. From primary school to high school, the number of students is gradually decreasing, which implies the influence of educational resources allocation, academic challenges or other factors on the student retention rate (Figure 5).

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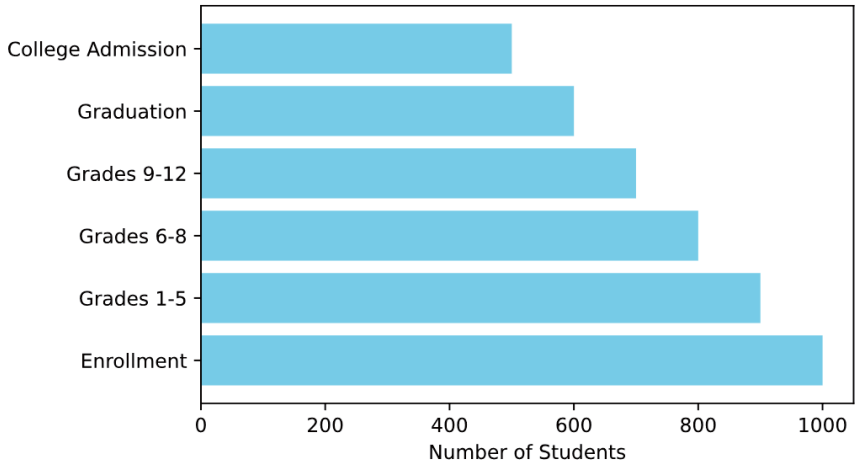


Fig. 5. Graduation rate and enrollment rate

At present, the evaluation of educational equity faces many problems. First of all, the evaluation standard is single, mainly focusing on academic performance, ignoring the multi-dimensional fairness of educational opportunities and processes. Secondly, the evaluation process is easily influenced by subjective factors, which leads to the objective and accurate results. In addition, the complexity and large-scale of educational data make it difficult to obtain and process data, which affects the accuracy and timeliness of evaluation. The causes of these problems include: imperfect evaluation system and lack of comprehensive and systematic evaluation standards; The data sharing mechanism is not perfect, which leads to the difficulty and low quality of data acquisition; As well as backward technical means, it is difficult to effectively handle large-scale complex data.

In view of the above problems, the educational equity evaluation model and optimization strategy recommendation algorithm based on computer algorithm proposed in this study provide new ideas and methods for scientific evaluation and effective optimization of educational equity. Through the distributed processing architecture of cloud computing platform, the efficient processing and analysis of large-scale educational data are realized; By constructing a multi-dimensional evaluation index system, it fully reflects the multi-dimensional characteristics of educational equity; By optimizing the strategy recommendation algorithm, targeted optimization suggestions are provided for different regions, schools or student groups. These achievements are of great significance for promoting the continuous improvement of educational equity.

IV. Optimization Strategies and Suggestions

A. Strategy presentation

In order to improve the fairness of education, this study proposes three optimization strategies based on computer algorithms. First, establish diversified evaluation standards, integrate multi-dimensional data through machine learning algorithms, and build a comprehensive student evaluation system; The second is to strengthen the fairness and transparency of education evaluation, adopt blockchain technology to ensure that data cannot be tampered with, and use smart contracts to automatically implement evaluation rules to reduce human intervention; Thirdly, the algorithm is used to optimize the allocation of educational resources, and the relationship between demand and allocation is analyzed through association rule mining and prediction model to predict future demand, and the optimization algorithm is used to formulate the optimal allocation scheme to ensure the balance and efficiency of resource allocation.

B. Implement recommendations

At the policy level, the government needs to introduce policies to support diversified evaluation standards, strengthen supervision of education evaluation, formulate policies to optimize resource allocation and set up special funds; In terms of technology, establish an educational big data platform, integrate data resources, introduce advanced AI technology, improve the intelligent level of evaluation and distribution, and ensure data security; In personnel training, educators are trained in diversified evaluation and algorithm application to improve their professional ability and social responsibility; In terms of social participation and supervision, we should establish a social participation mechanism, invite many parties to participate in education evaluation and resource allocation, set up complaint channels, and regularly optimize progress publicly to ensure transparency in the process and fair results.

V. Conclusion

In this study, a comprehensive education equity evaluation model is constructed, which integrates multi-source data such as students' academic performance, allocation of educational resources, teachers, school infrastructure and students' socio-economic background, and adopts cloud computing distributed processing architecture to ensure data processing efficiency. The evaluation model covers three categories of indicators: educational opportunity, process and fairness of results. The weights are allocated by combining expert scoring with machine learning to achieve comprehensive scoring. The research also designs an optimization strategy recommendation algorithm, and provides personalized optimization suggestions for different objects by using association rules mining. Empirical analysis shows that the traditional evaluation method has some problems, such as single standard and great subjective influence, and the evaluation model and optimization strategy based on computer algorithm proposed in this study can effectively solve these problems, which is of great value to improve educational fairness. The government,

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educational institutions and all walks of life should support the implementation of these strategies and jointly promote the development of education.

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